

AAYUSH KUMAR

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Education

The National Institute of Engineering

Bachelor of Engineering in Computer Science and Engineering (AI & ML)

Aug 2024 – May 2028

CGPA: 9.33 / 10

Technical Skills

Languages: C++, JavaScript (ES6+), TypeScript, Python, SQL, Java

Frontend: React.js, Vite, Tailwind CSS, HTML5, CSS3, Framer Motion

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, WebSockets

Databases: PostgreSQL, MongoDB, Redis, Firebase

DevOps & Tools: Docker, Linux (Fedora), Git, GitHub, GitHub Actions, CI/CD, Postman

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

Projects

Beatzy | *Python, FastAPI, React.js, Node.js, PostgreSQL, Redis, BullMQ, HuggingFace*

Apr 2026

- Designed a full-stack **music intelligence platform** with real-time audio analysis: BPM detection, key estimation, chord recognition, mood classification, and lyric synchronization.
- Built a **FastAPI microservice** using librosa, YAMNet, and scikit-learn for spectral feature extraction; integrated MusicBrainz and iTunes APIs for metadata enrichment and cover-art retrieval.
- Engineered a **scalable async pipeline** using BullMQ, Redis caching, and PostgreSQL persistence; deployed microservices on Render and Vercel with HTTP polling fallback.

LeadForge | *Python, SerpAPI, Apollo API, Crunchbase API, Data Pipelines*

Feb 2026

- Built an **automated B2B lead-generation pipeline** processing 500+ enrichment workflows; aggregated hiring signals, funding data, and company intelligence to cut manual research effort.
- Integrated **SerpAPI, Apollo, and Crunchbase APIs** for multi-source aggregation with validation, deduplication, enrichment, and lead-scoring for targeted outbound campaigns.

DeskGuard | *React.js, Node.js, Express.js, PostgreSQL, Redis, SSE, Docker*

May 2026

- Built a full-stack **library seat booking and anti-hoarding platform** featuring an interactive live desk map, QR-based check-ins, and automated seat reclamation workflows.
- Engineered a **real-time synchronization system** using **Server-Sent Events (SSE)** enabling millisecond-level desk status updates across all connected clients without polling.
- Designed a **hybrid persistence architecture** with PostgreSQL for audit logs and Redis TTL timers for session management, while automating abandoned-seat recovery through scheduled background sweep jobs.

CivicResolve | *React.js, Vite, Tailwind CSS, REST APIs, JWT*

Mar 2026

- Engineered a **civic grievance platform** for issue reporting, complaint tracking, and department-level workflow management for administrators and citizens.
- Built a reusable **React component library** with JWT-authenticated CRUD and responsive dashboards with real-time REST API polling for live status updates.

Experience

OWASP NIE Student Chapter

Aug 2025 – Present

Technical Core Team Member

NIE, Mysore

- Managed **technical infrastructure and CI/CD workflows** for campus hackathons and developer events, supporting 100+ participants per event.
- Delivered Git/GitHub workshops on branching and open-source workflows; automated event registration and participant management pipelines.

Achievements

- **Hackathon:** Secured **3rd place** at ThinkRoot × Vortex'26 (NIT Trichy) — built LeadForge in 24 hours, competing against 100+ teams.
- **Competitive Programming:** Solved **500+ DSA problems** across platforms on Graphs, Dynamic Programming, Trees, and Advanced Data Structures.
- **Open Source:** Active contributor to GitHub; maintains a portfolio of production-grade full-stack and ML applications.